

Quiz 4.8

Write a MATLAB function `t=t2rv(m)` that generates m samples of a random variable with the PDF $f_{T|T>2}(t)$ as given in Example 7.10.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

4.2.1 ● The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0 & x < -1, \\ (x+1)/2 & -1 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

- (a) What is $P[X > 1/2]$?
- (b) What is $P[-1/2 < X \leq 3/4]$?
- (c) What is $P[|X| \leq 1/2]$?
- (d) What is the value of a such that $P[X \leq a] = 0.8$?

4.2.2 ● The CDF of the continuous random variable V is

$$F_V(v) = \begin{cases} 0 & v < -5, \\ c(v+5)^2 & -5 \leq v < 7, \\ 1 & v \geq 7. \end{cases}$$

- (a) What is c ?
- (b) What is $P[V > 4]$?
- (c) What is $P[-3 < V \leq 0]$?
- (d) What is the value of a such that $P[V > a] = 2/3$?

4.2.3 ● In this problem, we verify that $\lim_{n \rightarrow \infty} \lceil nx \rceil / n = x$.

- (a) Verify that $nx \leq \lceil nx \rceil \leq nx + 1$.
- (b) Use part (a) to show

$$\lim_{n \rightarrow \infty} \lceil nx \rceil / n = x.$$

- (c) Use a similar argument to show that $\lim_{n \rightarrow \infty} \lfloor nx \rfloor / n = x$.

4.2.4 ● The CDF of random variable W is

$$F_W(w) = \begin{cases} 0 & w < -5, \\ \frac{w+5}{8} & -5 \leq w < -3, \\ \frac{1}{4} & -3 \leq w < 3, \\ \frac{1}{4} + \frac{3(w-3)}{8} & 3 \leq w < 5, \\ 1 & w \geq 5. \end{cases}$$

- (a) What is $P[W \leq 4]$?
- (b) What is $P[-2 < W \leq 2]$?
- (c) What is $P[W > 0]$?
- (d) What is the value of a such that $P[W \leq a] = 1/2$?

4.3.1 ● The random variable X has probability density function

$$f_X(x) = \begin{cases} cx & 0 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Use the PDF to find

- (a) the constant c ,
- (b) $P[0 \leq X \leq 1]$,
- (c) $P[-1/2 \leq X \leq 1/2]$,
- (d) the CDF $F_X(x)$.

4.3.2 ● The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0 & x < -1, \\ (x+1)/2 & -1 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

Find the PDF $f_X(x)$ of X .

4.3.3 ● Find the PDF $f_U(u)$ of the random variable U in Problem 4.2.4.

4.3.4 ■ For a constant parameter $a > 0$, a Rayleigh random variable X has PDF

$$f_X(x) = \begin{cases} a^2 x e^{-a^2 x^2/2} & x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

What is the CDF of X ?

4.3.5 ♦ Random variable X has a PDF of the form $f_X(x) = \frac{1}{2}f_1(x) + \frac{1}{2}f_2(x)$, where

$$f_1(x) = \begin{cases} c_1 & 0 \leq x \leq 2, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_2(x) = \begin{cases} c_2 e^{-x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

What conditions must c_1 and c_2 satisfy so that $f_X(x)$ is a valid PDF?

4.3.6 ♦♦ For constants a and b , random variable X has PDF

$$f_X(x) = \begin{cases} ax^2 + bx & 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

What conditions on a and b are necessary and sufficient to guarantee that $f_X(x)$ is a valid PDF?

4.4.1 ● Random variable X has PDF

$$f_X(x) = \begin{cases} 1/4 & -1 \leq x \leq 3, \\ 0 & \text{otherwise.} \end{cases}$$

Define the random variable Y by $Y = h(X) = X^2$.

- Find $E[X]$ and $\text{Var}[X]$.
- Find $h(E[X])$ and $E[h(X)]$.
- Find $E[Y]$ and $\text{Var}[Y]$.

4.4.2 ● Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 1/8 & 1 \leq x \leq 9, \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y = h(X) = 1/\sqrt{X}$.

- Find $E[X]$ and $\text{Var}[X]$.
- Find $h(E[X])$ and $E[h(X)]$.
- Find $E[Y]$ and $\text{Var}[Y]$.

4.4.3 ● Random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < 0, \\ x/2 & 0 \leq x \leq 2, \\ 1 & x > 2. \end{cases}$$

- What is $E[X]$?
- What is $\text{Var}[X]$?

4.4.4 ● The probability density function of random variable Y is

$$f_Y(y) = \begin{cases} y/2 & 0 \leq y < 2, \\ 0 & \text{otherwise.} \end{cases}$$

What are $E[Y]$ and $\text{Var}[Y]$?

4.4.5 ● The cumulative distribution function of the random variable Y is

$$F_Y(y) = \begin{cases} 0 & y < -1, \\ (y+1)/2 & -1 \leq y \leq 1, \\ 1 & y > 1. \end{cases}$$

What are $E[Y]$ and $\text{Var}[Y]$?

4.4.6 ■ The cumulative distribution function of random variable V is

$$F_V(v) = \begin{cases} 0 & v < -5, \\ (v+5)^2/144 & -5 \leq v < 7, \\ 1 & v \geq 7. \end{cases}$$

- What are $E[V]$ and $\text{Var}[V]$?
- What is $E[V^3]$?

4.4.7 ■ The cumulative distribution function of random variable U is

$$F_U(u) = \begin{cases} 0 & u < -5, \\ \frac{u+5}{8} & -5 \leq u < -3, \\ \frac{1}{4} & -3 \leq u < 3, \\ \frac{3u-7}{8} & 3 \leq u < 5, \\ 1 & u \geq 5. \end{cases}$$

- What are $E[U]$ and $\text{Var}[U]$?
- What is $E[2^U]$?

4.4.8 ■ X is a Pareto (α, μ) random variable, as defined in Appendix A. What is the largest value of n for which the n th moment $E[X^n]$ exists? For all feasible values of n , find $E[X^n]$.

4.5.1 ● Y is a continuous uniform $(1, 5)$ random variable.

- (a) What is $P[Y > E[Y]]$?
- (b) What is $P[Y \leq \text{Var}[Y]]$?

4.5.2 ● The current Y across a $1 \text{ k}\Omega$ resistor is a continuous uniform $(-10, 10)$ random variable. Find $P[|Y| < 3]$.

4.5.3 ● Radars detect flying objects by measuring the power reflected from them. The reflected power of an aircraft can be modeled as a random variable Y with PDF

$$f_Y(y) = \begin{cases} \frac{1}{P_0} e^{-y/P_0} & y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where $P_0 > 0$ is some constant. The aircraft is correctly identified by the radar if the reflected power of the aircraft is larger than its average value. What is the probability $P[C]$ that an aircraft is correctly identified?

4.5.4 ● Y is an exponential random variable with variance $\text{Var}[Y] = 25$.

- (a) What is the PDF of Y ?
- (b) What is $E[Y^2]$?
- (c) What is $P[Y > 5]$?

4.5.5 ● The time delay Y (in milliseconds) that your computer needs to connect to an access point is an exponential random variable.

- (a) Find $P[Y > E[Y]]$.
- (b) Find $P[Y > 2E[Y]]$.

4.5.6 ● X is an Erlang (n, λ) random variable with parameter $\lambda = 1/3$ and expected value $E[X] = 15$.

- (a) What is the value of the parameter n ?
- (b) What is the PDF of X ?
- (c) What is $\text{Var}[X]$?

4.5.7 ● Y is an Erlang $(n = 2, \lambda = 2)$ random variable.

- (a) What is $E[Y]$?
- (b) What is $\text{Var}[Y]$?
- (c) What is $P[0.5 \leq Y < 1.5]$?

4.5.8 ■ U is a zero mean continuous uniform random variable. What is $P[U^2 \leq \text{Var}[U]]$?

4.5.9 ■ U is a continuous uniform random variable such that $E[U] = 10$ and $P[U > 12] = 1/4$. What is $P[U < 9]$?

4.5.10 ■ X is a continuous uniform $(-5, 5)$ random variable.

- (a) What is the PDF $f_X(x)$?
- (b) What is the CDF $F_X(x)$?
- (c) What is $E[X]$?
- (d) What is $E[X^5]$?
- (e) What is $E[e^X]$?

4.5.11 ■ X is a continuous uniform $(-a, a)$ random variable. Find $P[|X| \leq \text{Var}[X]]$.

4.5.12 ■ X is a uniform random variable with expected value $\mu_X = 7$ and variance $\text{Var}[X] = 3$. What is the PDF of X ?

4.5.13 ■ The probability density function of random variable X is

$$f_X(x) = \begin{cases} (1/2)e^{-x/2} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $P[1 \leq X \leq 2]$?
- (b) What is $F_X(x)$, the cumulative distribution function of X ?
- (c) What is $E[X]$, the expected value of X ?
- (d) What is $\text{Var}[X]$, the variance of X ?

4.5.14 ■ Verify parts (b) and (c) of Theorem 4.6 by directly calculating the expected value and variance of a uniform random variable with parameters $a < b$.

4.5.15 ■ Long-distance calling plan A offers flat-rate service at 10 cents per minute. Calling plan B charges 99 cents for every call under 20 minutes; for calls over 20 minutes, the charge is 99 cents for the first 20 minutes plus 10 cents for every additional

minute. (Note that these plans measure your call duration exactly, without rounding to the next minute or even second.) If your long-distance calls have exponential distribution with expected value τ minutes, which plan offers a lower expected cost per call?

4.5.16 ■ In this problem we verify that an Erlang (n, λ) PDF integrates to 1. Let the integral of the n th order Erlang PDF be denoted by

$$I_n = \int_0^\infty \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!} dx.$$

First, show directly that the Erlang PDF with $n = 1$ integrates to 1 by verifying that $I_1 = 1$. Second, use integration by parts (Appendix B, Math Fact B.10) to show that $I_n = I_{n-1}$.

4.5.17 ■ Calculate the k th moment $E[X^k]$ of an Erlang (n, λ) random variable X . Use your result to verify Theorem 4.10. Hint: Remember that the Erlang $(n + k, \lambda)$ PDF integrates to 1.

4.5.18 ♦ In this problem, we outline the proof of Theorem 4.11.

- (a) Let X_n denote an Erlang (n, λ) random variable. Use the definition of the Erlang PDF to show that for any $x \geq 0$,

$$F_{X_n}(x) = \int_0^x \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!} dt.$$

- (b) Apply integration by parts (see Appendix B, Math Fact B.10) to this integral to show that for $x \geq 0$,

$$F_{X_n}(x) = F_{X_{n-1}}(x) - \frac{(\lambda x)^{n-1} e^{-\lambda x}}{(n-1)!}.$$

- (c) Use the fact that $F_{X_1}(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ to verify the claim of Theorem 4.11.

4.5.19 ♦ Prove by induction that an exponential random variable X with expected

value $1/\lambda$ has n th moment

$$E[X^n] = \frac{n!}{\lambda^n}.$$

Hint: Use integration by parts (Appendix B, Math Fact B.10).

4.5.20 ♦♦ This problem outlines the steps needed to show that a nonnegative continuous random variable X has expected value

$$E[X] = \int_0^\infty [1 - F_X(x)] dx.$$

- (a) For any $r \geq 0$, show that

$$r P[X > r] \leq \int_r^\infty x f_X(x) dx.$$

- (b) Use part (a) to argue that if $E[X] < \infty$, then

$$\lim_{r \rightarrow \infty} r P[X > r] = 0.$$

- (c) Now use integration by parts (Appendix B, Math Fact B.10) to evaluate

$$\int_0^\infty [1 - F_X(x)] dx.$$

4.6.1 ● The peak temperature T , as measured in degrees Fahrenheit, on a July day in New Jersey is the Gaussian $(85, 10)$ random variable. What is $P[T > 100]$, $P[T < 60]$, and $P[70 \leq T \leq 100]$?

4.6.2 ● What is the PDF of Z , the standard normal random variable?

4.6.3 ● Find each probability.

- (a) V is a Gaussian $(\mu = 0, \sigma = 2)$ random variable. Find $P[V > 4]$.
- (b) W is a Gaussian $(\mu = 2, \sigma = 5)$ random variable. What is $P[W \leq 2]$?
- (c) For a Gaussian $(\mu, \sigma = 2)$ random variable X , find $P[X \leq \mu + 1]$.
- (d) Y is a Gaussian $(\mu = 50, \sigma = 10)$ random variable. Calculate $P[Y > 65]$.

4.6.4 ● In each of the following cases, Y is a Gaussian random variable. Find the expected value $\mu = E[Y]$.

- (a) Y has standard deviation $\sigma = 10$ and $P[Y \leq 10] = 0.933$.
- (b) Y has standard deviation $\sigma = 10$ and $P[Y \leq 0] = 0.067$.
- (c) Y has standard deviation σ and $P[Y \leq 10] = 0.977$. (Find μ as a function of σ .)
- (d) $P[Y > 5] = 1/2$.

4.6.5 ● Your internal body temperature T in degrees Fahrenheit is a Gaussian ($\mu = 98.6, \sigma = 0.4$) random variable. In terms of the $\Phi(\cdot)$ function, find $P[T > 100]$. Does this model seem reasonable?

4.6.6 ● The temperature T in this thermostatically controlled lecture hall is a Gaussian random variable with expected value $\mu = 68$ degrees Fahrenheit. In addition, $P[T < 66] = 0.1587$. What is the variance of T ?

4.6.7 ● X is a Gaussian random variable with $E[X] = 0$ and $P[|X| \leq 10] = 0.1$. What is the standard deviation σ_X ?

4.6.8 ● A function commonly used in communications textbooks for the tail probabilities of Gaussian random variables is the complementary error function, defined as

$$\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^\infty e^{-x^2} dx.$$

Show that

$$Q(z) = \frac{1}{2} \operatorname{erfc}\left(\frac{z}{\sqrt{2}}\right).$$

4.6.9 ■ The peak temperature T , in degrees Fahrenheit, on a July day in Antarctica is a Gaussian random variable with a variance of 225. With probability $1/2$, the temperature T exceeds -75 degrees. What is $P[T > 0]$? What is $P[T < -100]$?

4.6.10 ■ A professor pays 25 cents for each blackboard error made in lecture to the student who points out the error. In a career

of n years filled with blackboard errors, the total amount in dollars paid can be approximated by a Gaussian random variable Y_n with expected value $40n$ and variance $100n$. What is the probability that Y_{20} exceeds 1000? How many years n must the professor teach in order that $P[Y_n > 1000] > 0.99$?

4.6.11 ■ Suppose that out of 100 million men in the United States, 23,000 are at least 7 feet tall. Suppose that the heights of U.S. men are independent Gaussian random variables with a expected value of $5'10''$. Let N equal the number of men who are at least $7'6''$ tall.

- (a) Calculate σ_X , the standard deviation of the height of U.S. men.
- (b) In terms of the $\Phi(\cdot)$ function, what is the probability that a randomly chosen man is at least 8 feet tall?
- (c) What is the probability that no man alive in the United States today is at least $7'6''$ tall?
- (d) What is $E[N]$?

4.6.12 ■ In this problem, we verify that for $x \geq 0$,

$$\Phi(x) = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right).$$

- (a) Let Y have a Gaussian $(0, 1/\sqrt{2})$ distribution and show that

$$F_Y(y) = \int_{-\infty}^y f_Y(u) du = \frac{1}{2} + \operatorname{erf}(y).$$

- (b) Observe that $Z = \sqrt{2}Y$ is Gaussian $(0, 1)$ and show that

$$\Phi(z) = F_Z(z) = F_Y\left(\frac{z}{\sqrt{2}}\right).$$

4.6.13 ■ This problem outlines the steps needed to show that the Gaussian PDF integrates to unity. For a Gaussian (μ, σ) random variable W , we will show that

$$I = \int_{-\infty}^{\infty} f_W(w) dw = 1.$$

- (a) Use the substitution $x = (w - \mu)/\sigma$ to show that

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} dx.$$

- (b) Show that

$$I^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy.$$

- (c) Change to polar coordinates to show that $I^2 = 1$.

4.6.14 ♦ At time $t = 0$, the price of a stock is a constant k dollars. At time $t > 0$ the price of a stock is a Gaussian random variable X with $E[X] = k$ and $\text{Var}[X] = t$. At time t , a *Call Option at Strike k* has value $V = (X - k)^+$, where the operator $(\cdot)^+$ is defined as $(z)^+ = \max(z, 0)$.

- (a) Find the expected value $E[V]$.
- (b) Suppose you can buy the call option for d dollars at time $t = 0$. At time t , you can sell the call for V dollars and earn a profit (or loss perhaps) of $R = V - d$ dollars. Let d_0 denote the value of d such that $P[R > 0] = 1/2$. Your strategy is that you buy the option if $d \leq d_0$ so that your probability of a profit is $P[R > 0] \geq 1/2$. Find d_0 .
- (c) Let d_1 denote the value of d such that $E[R] = 0.01 \times d$. Now your strategy is to buy the option if $d \leq d_1$ so that your expected return is at least one percent of the option cost. Find d_1 .
- (d) Are the strategies “Buy the option if $d \leq d_0$ ” and “Buy the option if $d \leq d_1$ ” reasonable strategies?

4.6.15 ♦ In mobile radio communications, the radio channel can vary randomly. In particular, in communicating with a fixed transmitter power over a “Rayleigh fading” channel, the receiver signal-to-noise ratio Y is an exponential random variable with expected value γ . Moreover, when $Y = y$, the probability of an error in decoding a transmitted bit is $P_e(y) = Q(\sqrt{2y})$ where $Q(\cdot)$ is the standard normal complementary CDF.

The average probability of bit error, also known as the bit error rate or BER, is

$$\bar{P}_e = E[P_e(Y)] = \int_{-\infty}^{\infty} Q(\sqrt{2y}) f_Y(y) dy.$$

Find a simple formula for the BER $\bar{P}_e$ as a function of the average SNR γ .

4.6.16 ♦♦ At time $t = 0$, the price of a stock is a constant k dollars. At some future time $t > 0$, the price X of the stock is a uniform $(k - t, k + t)$ random variable. At this time t , a *Put Option at Strike k* (which is the right to sell the stock at price k) has value $(k - X)^+$ dollars where the operator $(\cdot)^+$ is defined as $(z)^+ = \max(z, 0)$. Similarly a *Call Option at Strike k* (the right to buy the stock at price k) at time t has value $(X - k)^+$.

- (a) At time 0, you sell the put and receive d dollars. At time t , you purchase the put for $(k - X)^+$ dollars to cancel your position. Your gain is

$$R = g_p(X) = d - (k - X)^+.$$

Find the central moments $E[R]$ and $\text{Var}[R]$.

- (b) In a short straddle, you sell the put for d dollars and you also sell the call for d dollars. At a future time $t > 0$, you purchase the put for $(k - X)^+$ dollars and the call for $(X - k)^+$ dollars to cancel both positions. Your gain on the put is $g_p(X) = d - (k - X)^+$ dollars and your gain on the call is $g_c(X) = d - (X - k)^+$ dollars. Your net gain is

$$R' = g_p(X) + g_c(X).$$

Find the expected value $E[R']$ and variance $\text{Var}[R']$.

- (c) Explain why selling the straddle might be attractive compared to selling just the put or just the call.

4.6.17 ♦♦ Continuing Problem 4.6.16, suppose you sell the straddle at time $t = 0$ and liquidate your position at time t , generating a profit (or perhaps a loss) R' . Find the

PDF $f_{R'}(r)$ of R' . Suppose d is sufficiently large that $E[R'] > 0$. Would you be interested in selling the short straddle? Are you getting something, namely $E[R']$ dollars, for nothing?

4.7.1 ● Let X be a random variable with CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ x/3 + 1/3 & -1 \leq x < 0, \\ x/3 + 2/3 & 0 \leq x < 1, \\ 1 & 1 \leq x. \end{cases}$$

Sketch the CDF and find

- (a) $P[X < -1]$ and $P[X \leq -1]$,
- (b) $P[X < 0]$ and $P[X \leq 0]$,
- (c) $P[0 < X \leq 1]$ and $P[0 \leq X \leq 1]$.

4.7.2 ● Let X be a random variable with CDF

$$F_X(x) = \begin{cases} 0 & x < -1, \\ x/4 + 1/2 & -1 \leq x < 1, \\ 1 & 1 \leq x. \end{cases}$$

Sketch the CDF and find

- (a) $P[X < -1]$ and $P[X \leq -1]$.
- (b) $P[X < 0]$ and $P[X \leq 0]$.
- (c) $P[X > 1]$ and $P[X \geq 1]$.

4.7.3 ● For random variable X of Problem 4.7.2, find $f_X(x)$, $E[X]$, and $\text{Var}[X]$.

4.7.4 ● X is Bernoulli random variable with expected value p . What is the PDF $f_X(x)$?

4.7.5 ● X is a geometric random variable with expected value $1/p$. What is the PDF $f_X(x)$?

4.7.6 ■ When you make a phone call, the line is busy with probability 0.2 and no one answers with probability 0.3. The random variable X describes the conversation time (in minutes) of a phone call that is answered. X is an exponential random variable with $E[X] = 3$ minutes. Let the random variable W denote the conversation time (in seconds) of all calls ($W = 0$ when the line is busy or there is no answer.)

- (a) What is $F_W(w)$?
- (b) What is $f_W(w)$?
- (c) What are $E[W]$ and $\text{Var}[W]$?

4.7.7 ■ For 80% of lectures, Professor X arrives on time and starts lecturing with delay $T = 0$. When Professor X is late, the starting time delay T is uniformly distributed between 0 and 300 seconds. Find the CDF and PDF of T .

4.7.8 ♦ With probability 0.7, the toss of an Olympic shot-putter travels $D = 60 + X$ feet, where X is an exponential random variable with expected value $\mu = 10$. Otherwise, with probability 0.3, a foul is committed by stepping outside of the shot-put circle and we say $D = 0$. What are the CDF and PDF of random variable D ?

4.7.9 ♦ For 70% of lectures, Professor Y arrives on time. When Professor Y is late, the arrival time delay is a continuous random variable uniformly distributed from 0 to 10 minutes. Yet, as soon as Professor Y is 5 minutes late, all the students get up and leave. (It is unknown if Professor Y still conducts the lecture.) If a lecture starts when Professor Y arrives and always ends 80 minutes after the scheduled starting time, what is the PDF of T , the length of time that the students observe a lecture.

4.8.1 ● Write a function `y=quiz31rv(m)` that produces m samples of random variable Y defined in Quiz 4.2.

4.8.2 ● For the Gaussian $(0, 1)$ complementary CDF $Q(z)$, a useful numerical approximation for $z \geq 0$ is

$$\hat{Q}(z) = \left(\sum_{n=1}^5 a_n t^n \right) e^{-z^2/2},$$

where

$$\begin{aligned} t &= \frac{1}{1 + 0.231641888z} & a_1 &= 0.127414796 \\ a_2 &= -0.142248368 & a_3 &= 0.7107068705 \\ a_4 &= -0.7265760135 & a_5 &= 0.5307027145. \end{aligned}$$

To compare this approximation to $Q(z)$, use MATLAB to graph

$$e(z) = \frac{Q(z) - \hat{Q}(z)}{Q(z)}.$$

4.8.3 ■ Use `exponentialrv.m` and Theorem 4.9 and to write a MATLAB function `k=georv(p,m)` that generates m samples of a geometric (p) random variable K . Compare the resulting algorithm to the technique employed in Problem 3.9.8 for `geometricrv(p,m)`.

4.8.4 ■ Applying Equation (4.14) with x replaced by $i\Delta$ and dx replaced by Δ , we obtain

$$P[i\Delta < X \leq i\Delta + \Delta] = f_X(i\Delta) \Delta.$$

If we generate a large number n of samples of random variable X , let n_i denote the number of occurrences of the event

$$\{i\Delta < X \leq (i+1)\Delta\}.$$

We would expect that

$$\lim_{n \rightarrow \infty} \frac{n_i}{n} = f_X(i\Delta) \Delta,$$

or equivalently,

$$\lim_{n \rightarrow \infty} \frac{n_i}{n\Delta} = f_X(i\Delta).$$

Use MATLAB to confirm this with $\Delta = 0.01$ for

- (a) an exponential ($\lambda = 1$) random variable X and for $i = 0, \dots, 500$,
 - (b) a Gaussian $(3, 1)$ random variable X and for $i = 0, \dots, 600$.
-